Source catalogue

References are organized by topic, then readable year/title folder names. ArXiv identifiers and versions remain in metadata. Legacy-name lookup maps the earlier identifiers to the current folders.

Collection status and code availability · Machine-readable manifest

Each paper folder contains its PDF, extracted text, available LaTeX source, metadata, and discovered code archives. A linked code archive may be a baseline or dataset dependency; inspect its recorded role before reproducing an experiment.

Scaling laws and fitting

Work	Identifier / source	Local files
Scaling Laws for Neural Language Models	2001.08361	Read · Metadata
Training Compute-Optimal Large Language Models (Chinchilla)	2203.15556	Read · Metadata
A Hitchhiker's Guide to Scaling Law Estimation	2410.11840v2	Read · Metadata
Beyond Chinchilla-Optimal: Accounting for Inference in Language Model Scaling Laws	2401.00448	Read · Metadata
Chinchilla Scaling: A Replication Attempt	2404.10102	Read · Metadata
Language models scale reliably with over-training and on downstream tasks	2403.08540v2	Read · Metadata
Resolving Discrepancies in Compute-Optimal Scaling of Language Models	2406.19146	Read · Metadata
Scaling Law with Learning Rate Annealing	2408.11029v2	Read · Metadata
Scaling Laws for Precision	2411.04330	Read · Metadata
Scaling Laws with Vocabulary: Larger Models Deserve Larger Vocabularies	2407.13623	Read · Metadata
Gemstones: A Model Suite for Multi-Faceted Scaling Laws	2502.06857	Read · Metadata
(Mis)Fitting: A Survey of Scaling Laws	2502.18969	Read · Metadata
Predictable Scale: Part II, Farseer: A Refined Scaling Law in Large Language Models	2506.10972v3	Read · Metadata
Skaling: Chinchilla's Exponents Meet Kaplan's Coupling	2608.07222v1	Read · Metadata

Architecture and MoE

Work	Identifier / source	Local files
GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints	2305.13245	Read · Metadata
Scaling Laws for Fine-Grained Mixture of Experts	2402.07871	Read · Metadata
Cost-Optimal Grouped-Query Attention for Long-Context Modeling	2503.09579	Read · Metadata
Joint MoE Scaling Laws: Mixture of Experts Can Be Memory Efficient	2502.05172	Read · Metadata
Parameters vs FLOPs: Scaling Laws for Optimal Sparsity for Mixture-of-Experts Language Models	2501.12370	Read · Metadata
SMELT: Scaling Laws for Compute-Matched MoE Looped Transformers	2609.01343v3	Read · Metadata

Optimizers and hyperparameter transfer

Work	Identifier / source	Local files
An Empirical Model of Large-Batch Training	1812.06162	Read · Metadata
Model Soups: Averaging Weights of Multiple Fine-tuned Models	2203.05482v3	Read · Metadata

Work	Identifier / source	Local files
Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer (μ-Transfer)	2203.03466v2	Read · Metadata
Stop Wasting My Time! Saving Days of ImageNet and BERT Training with Latest Weight Averaging (LAWA)	2209.14981	Read · Metadata
Tensor Programs VI: Feature Learning in Infinite-Depth Neural Networks	2310.02244	Read · Metadata
Depthwise Hyperparameter Transfer in Residual Networks: Dynamics and Scaling Limit	2309.16620	Read · Metadata
Early Weight Averaging meets High Learning Rates for LLM Pre-training	2306.03241	Read · Metadata
DeepSeek LLM: Scaling Open-Source Language Models with Longtermism	2401.02954	Read · Metadata
Scaling Optimal LR Across Token Horizons	2409.19913	Read · Metadata
Understanding Warmup-Stable-Decay Learning Rates: A River Valley Loss Landscape Perspective	2410.05192v3	Read · Metadata
Critical Batch Size Revisited: A Simple Empirical Approach to Large-Batch Language Model Training	2505.23971	Read · Metadata
Fantastic Pretraining Optimizers and Where to Find Them	2509.02046v2	Read · Metadata
Model Merging in Pre-training of Large Language Models	2505.12082v3	Read · Metadata
Muon is Scalable for LLM Training	2502.16982	Read · Metadata
Power Lines: Scaling Laws for Weight Decay and Batch Size in LLM Pre-training	2505.13738v2	Read · Metadata
Scaling and Transferability of Annealing Strategies in Large Language Model Training	2512.13705	Read · Metadata
Predictable Scale: Part I, Step Law	2503.04715v3	Read · Metadata
How to Allocate Your Tokens? Scaling Laws with Training Steps and Batch Size	2607.01487v1	Read · Metadata
Small-Scale Experiments: Are We There Yet?	2608.11859	Read · Metadata
Weight Decay Improves Language Model Plasticity	2602.11137v2	Read · Metadata

Data construction, mixtures, and repetition

Work	Identifier / source	Local files
Scaling Data-Constrained Language Models	2305.16264v5	Read · Metadata
To Repeat or Not To Repeat: Insights from Scaling LLM under Token-Crisis	2305.13230v2	Read · Metadata
UniMax: Fairer and More Effective Language Sampling for Large-Scale Multilingual Pretraining	2304.09151	Read · Metadata
Nemotron-CC: Transforming Common Crawl into a Refined Long-Horizon Pretraining Dataset	2412.02595	Read · Metadata
Larger Datasets Can Be Repeated More: A Theoretical Analysis of Multi-Epoch Scaling in Linear Regression	2511.13421v2	Read · Metadata
Decouple Searching from Training: Scaling Data Mixing via Model Merging for Large Language Model Pre-training (DeMix)	2602.00747	Read · Metadata
Olmix: A Framework for Data Mixing Throughout LM Development	2602.12237	Read · Metadata
Prescriptive Scaling Laws for Data Constrained Training	2605.01640	Read · Metadata
Scaling Laws for Mixture Pretraining Under Data Constraints	2605.12715	Read · Metadata

Evaluation and task prediction

Work	Identifier / source	Local files
GPT-4 Technical Report	2303.08774v6	Read · Metadata
Establishing Task Scaling Laws via Compute-Efficient Model Ladders (OLMo Task Ladder)	2412.04403	Read · Metadata
RULER: What's the Real Context Size of Your Long-Context Language Models?	2404.06654v3	Read · Metadata
Why Has Predicting Downstream Capabilities of Frontier AI Models with Scale Remained Elusive?	2406.04391	Read · Metadata
Unveiling Downstream Performance Scaling of LLMs: A Clustering-Based Perspective (COD)	2502.17262v4	Read · Metadata
Language Model Evaluation Harness: checkpoint evaluation	Web report	Read · Metadata

Model technical reports

Work	Identifier / source	Local files
DeepSeek-V3 Technical Report	2412.19437	Read · Metadata
Nemotron-4 15B Technical Report	2402.16819	Read · Metadata
Nemotron-4 340B Technical Report	2406.11704	Read · Metadata
OLMo: Accelerating the Science of Language Models	2402.00838v4	Read · Metadata
OLMoE: Open Mixture-of-Experts Language Models	2409.02060	Read · Metadata
Phi-4 Technical Report	2412.08905	Read · Metadata
The Llama 3 Herd of Models	2407.21783	Read · Metadata
GLM-4.5: Agentic, Reasoning, and Coding (ARC) Foundation Models	2508.06471	Read · Metadata
Kimi K2: Open Agentic Intelligence	2507.20534v2	Read · Metadata
Nemotron 3 Nano: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning	2512.20848	Read · Metadata
2 OLMo 2 Furious	2501.00656v3	Read · Metadata
OLMo 3	2512.13961v1	Read · Metadata
Qwen3 Technical Report	2505.09388	Read · Metadata
Kimi K2.5: Visual Agentic Intelligence	2602.02276	Read · Metadata
Kimi K3: Open Frontier Intelligence	2607.24653	Read · Metadata
Nemotron 3 Super: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning	2604.12374	Read · Metadata
Nemotron 3 Ultra: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning	2606.15007	Read · Metadata

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Olmo-core 3 technical report	OLMo-core-3-technical-report	Read · Metadata
On the Design of Qwen3.8-Next Architecture: Evaluation, Efficiency, and Training Stability	2608.30320v1	Read · Metadata

Marin reports and pipelines

Work	Identifier / source	Local files
Marin Datakit released mixture experiment records: audited snapshot	Web report	Read · Metadata
Delphi: An Open Scaling Suite	Web report	Read · Metadata
Marin replication evidence: pinned PRs, HF data/results, long context, SFT and evaluation	Web report	Read · Metadata
Marin Data Pipeline Overview	Web report	Read · Metadata
Marin Pretraining Efficiency	Web report	Read · Metadata
Marin replication code audit: optimizer searches, MoE ladder and Datakit	Web report	Read · Metadata

Research methodology

Work	Identifier / source	Local files
On Over-fitting in Model Selection and Subsequent Selection Bias	Web report	Read · Metadata
Democratizing Large-Scale Mixture-of-Experts Training with NVIDIA PyTorch Parallelism	Web report	Read · Metadata
KernelWiki: source-pinned knowledge engineering and validation practices	Web report	Read · Metadata

Training systems and source implementations

Work	Identifier / source	Local files
Megatron-LM: training, indexed data and MoE implementation	Web report	Read · Metadata
TorchTitan: training, dataset and evaluation implementation	Web report	Read · Metadata